

8086 Interrupt Structure

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What is an Interrupt?

- An interrupt is the method of processing the microprocessor by peripheral device(hardware interrupt).
- The meaning of ‘interrupts’ is to break the sequence of operation.
- An interrupt is used to cause a temporary halt in the execution of program.
- Microprocessor responds to the interrupt with an interrupt service routine, which is short program or subroutine that instructs the microprocessor on how to handle the interrupt.
- After executing ISR , the control is transferred back again to the main program

Types of Interrupts in 8086

An 8086 interrupt can come from any one of the three sources:

- **Hardware Interrupt**: One source is an external signal applied to the Non Maskable Interrupt (**NMI**) input pin or to the Interrupt (**INTR**) input pin.
- **Software Interrupt** : A second source of an interrupt is execution of the Interrupt instruction- **INT**
- **Interrupt due to internally generated error conditions (Referred to as Type or Exception)**: The third source of an interrupt is some error condition produced in the 8086 by the execution of an instruction . An example of this is the divide by zero error

(**Type - Intel or Exception - Motorola)

Hardware and Software Interrupt

- The two basic type of hardware interrupts are: maskable and non-maskable.
- Nonmaskable interrupt requires an immediate response by microprocessor, it usually used for serious circumstances like power failure.
- A maskable interrupt is an interrupt that the microprocessor can ignore depending upon some predetermined condition defined by status register.
- **Hardware and Software Interrupts:**

- The non-maskable interrupt is generated by an external device, through a rising edge on the NMI pin
- A maskable interrupt is generated by an external device, through a high logic level on the INTR pin (the external device has to specify the interrupt number).
- **Software interrupts:** Software interrupts (exceptions) using the INT instruction (followed by the interrupt number (type)).

Interrupt Vector table

- The 8086 provides a 256 entry interrupt vector table beginning at address 0:0 in memory.
- This is a 1KB table containing 256 entries 4-bytes each. Each entry in this table contains a segmented address that points at the interrupt service routine in memory.
- The lowest five types are dedicated to specific interrupts such as the divide by zero interrupt and the non maskable interrupt.
- The next 27 interrupt types, from 5 to 31 are High priority 3 reserved by Intel for use in future microprocessors.
- The upper 224 interrupt types, from 32 to 255, are available to use for hardware and software interrupts

- The first 1Kbyte of memory of 8086 (00000 to 003FF) is set aside as a table for storing the starting addresses of Interrupt Service Procedures (ISP).
- Since 4-bytes are required for storing starting addresses of ISPs, the table can hold 256 Interrupt procedures.
- The starting address of an ISP is often called the Interrupt Vector or Interrupt Pointer. Therefore the table is referred as Interrupt Vector Table. In this table, IP value is put in as low word of the vector & CS is put in high vector.

Interrupt Vector table

INTERRUPTS

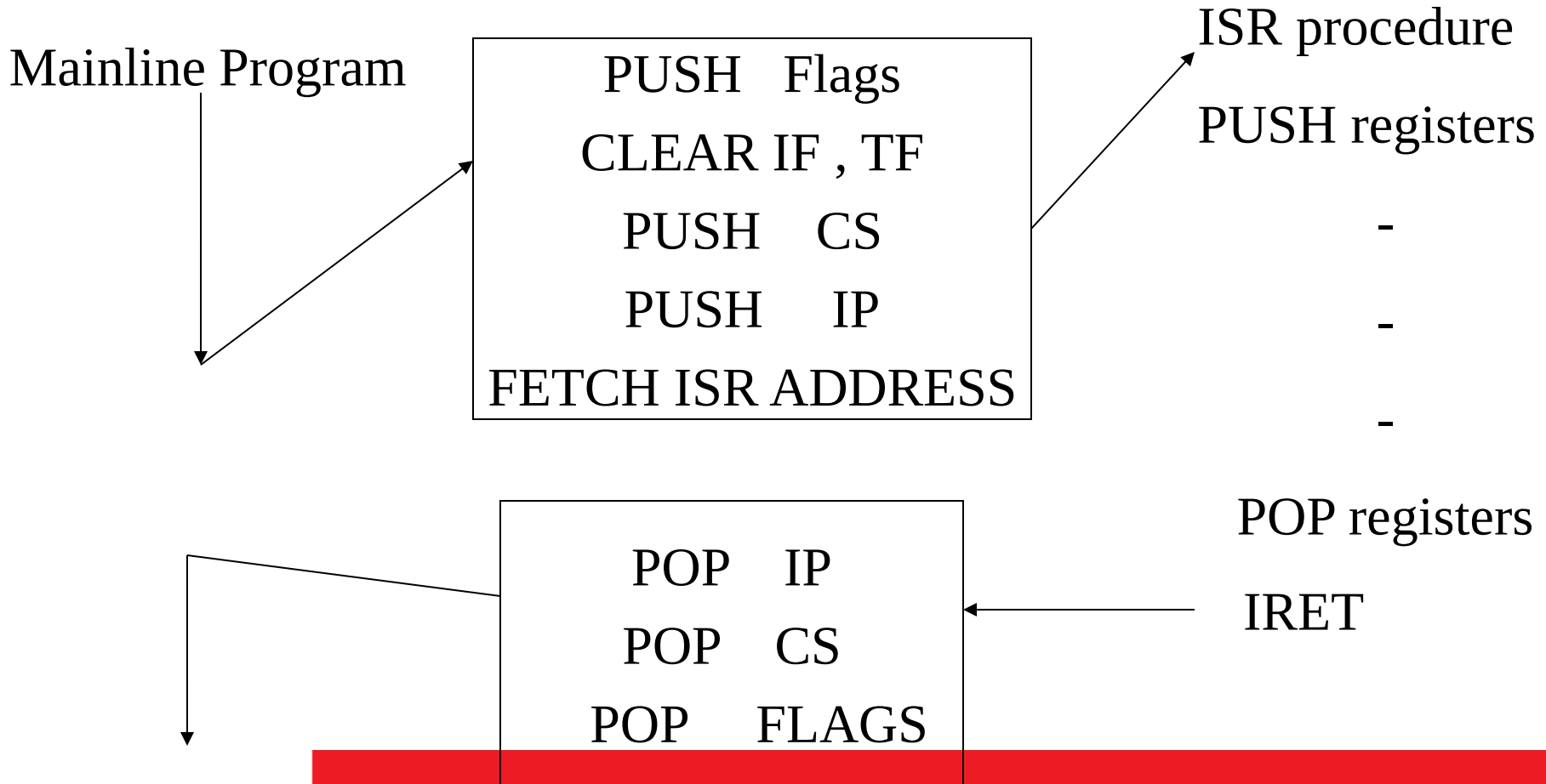
	Type 32 — 255 User interrupt vectors
080H	Type 14 — 31 Reserved
040H	Type 16 Coprocessor error
03CH	Type 15 Unassigned
038H	Type 14 Page fault
034H	Type 13 General protection
030H	Type 12 Stack segment overrun
02CH	Type 11 Segment not present
028H	Type 10 Invalid task state segment
024H	Type 9 Coprocessor segment overrun
020H	Type 8 Double fault
01CH	Type 7 Coprocessor not available
018H	Type 6 Undefined opcode
014H	Type 5 BOUND
010H	Type 4 Overflow (INTO)
00CH	Type 3 1-byte breakpoint
008H	Type 2 NMI pin
004H	Type 1 Single-step
000H	Type 0 Divide error

(a)

Interrupt priority

Interrupt	Priority
Divide Error, INT(n),INTO	Highest
NMI	↓
INTR	
Single Step	
	Lowest

Contd..



How does 8086 responds to the Interrupt

1. It decrements SP by 2 and pushes the flag register on the stack
2. Disables INTR by clearing the IF
3. It resets the TF in the flag Register
4. It decrements SP by 2 and pushes CS on the stack.
5. It decrements SP by 2 and pushes IP on the stack
6. Fetch the ISR address from the interrupt vector table

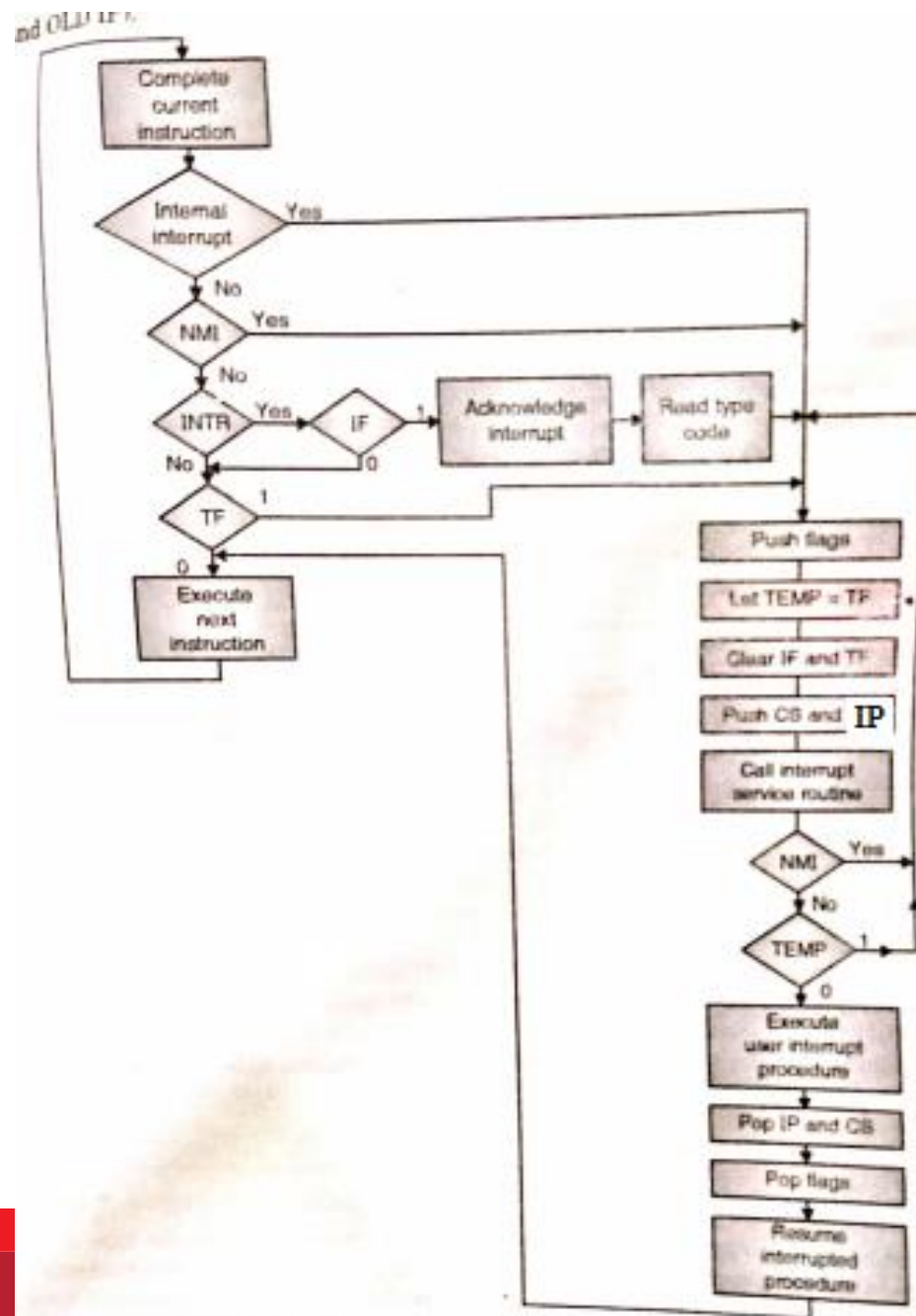


Fig. 7.3.1 : Interrupt procedure

Interrupt Vector Table

INT Number	Physical Address
INT 00	00000
INT 01	00004
INT 02	00008
:	:
:	:
INT FF	003FC

Physical Address of ISR for the particular Interrupt is obtained by multiplying the the type number by 4

Find the physical address in the interrupt vector table associated with

- a) INT 12H b) INT 8H

Solution: a) $12H * 4 = 48H$

Physical Address: 00048H (48 through 4BH are set aside for CS & IP)

b) $8 * 4 = 20H$

Memory Address : 00020H